

Alberto ROTA

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PROFILE

<i>About Me</i>	AI engineer and PhD researcher in computer vision and deep learning for surgical and medical imaging. My work spans 3D reconstruction, self-supervised representation learning, and dense point tracking, with a first-author CVPR 2026 Oral + Award Candidate ranking in the top 2% of submissions. I can lead a research agenda and take research from first idea to deployable, GPU-scaled pipeline and lead projects end to end.
<i>Objectives</i>	Be challenged · Solve puzzles · Make an impact

INDUSTRY EXPERIENCE

FEB 2026	AI Researcher - ASENSUS SURGICAL, NOW MERGED INTO KARL STORZ
FEB 2023	Focus: Computer Vision Deep Learning methods for enhancing the spatial and contextual informative content of endoscopic image data, with focus on 3D reconstruction and occlusion restoration - Developed, applied and surpassed state-of-the-art models and pipelines targeted at recovering 3D information from 2D endoscopic image data, with strong focus on self-supervised frameworks [NDA] - Researched, developed and tested geometry-aware learned representations of 3D endoscopic spaces and 2D images [NDA] - Worked in structured teams, both in contributing and leading positions - Gained project management, time management and DevOps skills

EDUCATION AND ACADEMIC RESEARCH

OCT 2026	Ph.D in Bioengineering - POLITECNICO DI MILANO & ASENSUS SURGICAL INC., MILAN, IT
FEB 2023	Thesis Title: <i>Self-Supervised Learning for Robust Visual Scene Understanding in Minimally Invasive Surgery: Toward Intelligent Intraoperative Assistance Without Domain-Specific Annotation</i>
DEC 2025	Visiting Research Fellow - CHAIR OF PHOTOGRAMMETRY AND REMOTE SENSING, TUM, MUNICH, DE
JULY 2025	Focus: Geometry-guided reconstruction for challenging visual domains.
SEP 2025	Teaching Assistant - NEARLAB MEDICAL ROBOTICS, POLITECNICO DI MILANO
SEP 2023	Primary Course: <i>Technologies for Motor Behavior Analysis and Virtual Modeling</i> Guest Lectures at: <i>Medical Robotics and Technologies for Computer Aided Surgery</i> Gained communication, didactic and public speaking skills
MAY 2026	Scientific Communicator - POLITECNICO DI MILANO
OCT 2023	Course: <i>Understanding Artificial Intelligence</i>  - Introduced the basics of neural networks and data-driven algorithms interactively to high school students, reaching out to over 16 schools and 500 students over the course of 2 years - Developed didactic and teaching skills
DEC 2022	MSc in Bioengineering - POLITECNICO DI MILANO, MILAN, IT
SEP 2020	Focus: AI and Computer Vision methods for 3D data in bioengineering; Virtualization of teleoperated surgical robotic environments

TECH STACK

<i>Languages</i>	Python (<i>expert</i>); C++, C#, C, MATLAB (<i>familiar</i>)
<i>DL / ML</i>	PyTorch (<i>expert</i>); Hugging Face, scikit-learn, WandB (<i>proficient</i>), CUDA (<i>hobby</i>)
<i>CV & 3D</i>	OpenCV, Open3D, Rerun (<i>expert</i>); Blender (<i>proficient</i>)
<i>MLOps</i>	Linux/Bash, Docker, Git, Slurm (<i>expert</i>); Multi-GPU/Distributed, Tailscale (<i>proficient</i>)
<i>Agentic AI</i>	OpenCode (<i>expert</i>): custom skills, subagents, hooks, MCP servers, slash commands, AGENTS.md context engineering; agent-assisted refactoring and research workflows (<i>proficient</i>)

RESEARCH PROJECT CONTRIBUTIONS

- JUN 2026 **UnReflectAnything [CVPR 2026 ORAL]** - PROJECT LEAD AND PRIMARY CONTRIBUTOR
JUL 2025 Conceived, developed, and led the full research agenda of *UnReflectAnything*, an RGB-only framework that removes specular highlights from arbitrary images by predicting a highlight map and reconstructing a reflection-free diffuse image. [1] 
- Designed a physically grounded synthetic ground-truth pipeline for compositing specular highlights onto any RGB image, and a token-space inpainting module that removes them at the feature level
 - Achieved competitive or state-of-the-art performance on the relevant benchmarks
 - Selected as CVPR Oral presentation and Award Candidate, ranking in the top 2% of submissions
- ONGOING **COAST: Causal Occlusion-Aware Surgical Point Tracking** - SOLE CONTRIBUTOR
MAY 2026 Reframed causal surgical point tracking as state estimation over a frozen tracker treated as a black-box sensor, with a learned inter-point motion prior carrying points through occlusions and a Kalman-like gate fusing prediction with measurement [3] 
- Built a passive state-space layer raising occlusion accuracy on three public trackers
 - Trained on synthetic data then adapted to unlabeled surgical video via induced occlusions, labeled by the agreement of two independent teacher trackers
- JUL 2025 **Self-Supervised Feature Matching in Endoscopic Surgery** - SOLE CONTRIBUTOR
DEC 2024 Developed an end-to-end self-supervised pipeline based on novel-view synthesis and contrastive optimization for semantic embedding adaptation of DINOv2 features towards a pixel matching task in the surgical endoscopy domain [4]
- Ideated and developed an SSL pipeline for establishing pseudo-ground-matches in source-synthetic endoscopic image pairs to be used for contrastive learning
 - Trained an adapter for DINOv2 to produce localized semantics, and surpassed state-of-the-art models on pixel matching tasks on surgical endoscopy imagery
- ONGOING **SpectraBREAST** - COMPUTER VISION PIPELINE LEADER
MAR 2026 Leading the research agenda, pipeline development and deployment for the computer vision workpackage of the EU-funded SpectraBREAST project, including 3D reconstruction tasks and Hyperspectral registration task. [5] 
- Successfully deployed a 3D reconstruction and registration framework for reconstructing hyperspectral 3D geometry from scans of *ex-vivo* breast lumpectomy specimens
 - Supervised a team of 2 people, collaborating with multi-disciplinary teams
- JUL 2022 **μ VES - microVascular Evaluation System** - LEADER AND PRIMARY CONTRIBUTOR
MAR 2020 Built a fully automated pipeline for the topological and morphological analysis of 3D micro-vascular networks images from confocal microscopy, with Deep-Learning-based confocal image segmentation and integration with a CFD simulation software [6] 
- Built and trained a 3D U-Net for confocal image segmentation, within a complete pipeline for quantitative analysis inclusive of skeletonization and morphological measurements
- DEC 2020 **STEVE - Surgical Training Enhanced Virtual Environment** - SOLE CONTRIBUTOR
FEB 2022 Built a virtual reality training environment targeting teleoperated surgical robotics, enhanced with visuo-haptic assistance-as-needed guidance, personalized adaptive difficulty and visual feedback for haptic force training [7] 
- Built a VR simulator for surgical robotics in C# with Unity, connected via ROS to a teleoperation console, with haptic assistance-as-needed guidance algorithms validated in a statistical study
- MAR 2025 **OVIT - Ovarian Cancer Resectability Prediction** - MINOR CONTRIBUTOR
OCT 2024 Participated in the development of a Deep Learning decision-support-system for Ovarian Cancer treatment planning [8] 
- Contributed to the development, statistical validation and academic publication of the pipeline
 - Designed and assembled a heavy-duty multi-GPU workstation for the clinical Deep Learning workload, with remote SSH/VNC access, Dockerized environments, and strict adherence to GDPR and data privacy best practices

OPEN SOURCE TOOLS

- DEC 2024 **Ground Control** - MAINTAINER
A Terminal-based package for monitoring system hardware in real time with rich plots and graphics in the terminal. Aimed for multi-GPU machines and ML workflows.  & PyPI
- AUG 2026 **Sekrt** - MAINTAINER
A secret manager for developers and DevOps engineers. Like **pass**, but with a modern interface, first-class API key, SSH keypair and .env file support, and painless sync through any private git remote  & PyPI
- APR 2025 **DaSSHboard** - MAINTAINER
A stylish customizable VS Code extension to manage multiple SSH remote connections with a smart one-click dashboard for faster access to remote development.  & 
- JUL 2025 **MedView** - CONTRIBUTOR
A fast environment to preview and interact with medical images directly from a code editor, with support for .nii and .dcm files, aimed at streamlining AI development tasks in the medical image analysis field.  & 

AWARDS

- JUN 2026 **Best Poster Award** - HAMLYN WORKSHOP ON PHYSICS- AND AI-DRIVEN DIGITAL TWINS
Coverage-Optimal Robotic Colonoscopy via Patient-Specific Digital-Twin World Models
- JUN 2023 **Best Application Award** - HAMLYN SURGICAL ROBOTICS CHALLENGE 2023
Haptic assistance for improving skill transfer in surgical robotics training 
- APR 2022 **Best Development Award** - POLIMI CAPSTONE PROJECTS 2022
SPINTEST - Data-Driven Compliancy Assessment for Extra-Corporeal Centrifugal Blood Pumps 

SELECTED RESEARCH PAPERS

- [1] **Alberto Rota**, Mert Kiray, Mert Asim Karaoglu, Patrick Rukhamp, Elena De Momi, Nassir Navab, and Benjamin Busam. UnReflectAnything: RGB-only highlight removal by rendering synthetic specular supervision. *Computer Vision and Pattern Recognition - CVPR*, 2026 .
- [2] Francesca Fati, **Alberto Rota**, Adriana V. Gregory, Anna Catozzo, Maria C. Giuliano, Mrinal Dhar, Luigi De Vitis, Annie T. Packard, Francesco Multinu, Elena De Momi, Carrie L. Langstraat, and Timothy L. Kline. Adapting foundation models for annotation-efficient adnexal mass segmentation in cine images. *CVPR Workshop on Subtle Visual Computing*, 2026 .
- [3] **Alberto Rota** and Elena De Momi. Self-supervised contrastive embedding adaptation for endoscopic image matching. *IEEE Transaction on Medical Imaging*, 2025 - Under Review .
- [4] **Alberto Rota** and Elena De Momi. Self-supervised contrastive embedding adaptation for endoscopic image matching. *IEEE Transaction on Medical Imaging*, 2025 - Under Review .
- [5] Anna Bicchi*, **Alberto Rota***, Leonardo Passoni*, Nicola Ancellotti, Andrea Peroni, Lorenzo Vinco, Dario Polli, and Elena De Momi. A lightweight fiducial-based pipeline for 3d hyperspectral mapping of ex-vivo lumpectomy specimens. *Ital-IA*, 2026 .
- [6] **Alberto Rota**, Luca Possenti, Giovanni S Offeddu, Martina Senesi, Adelaide Stucchi, Irene Venturelli, Tiziana Rancati, Paolo Zunino, Roger D Kamm, and Maria Laura Costantino. A three-dimensional method for morphological analysis and flow velocity estimation in microvasculature on-a-chip. *Bioengineering & Translational Medicine*, 2023 .
- [7] **Alberto Rota**, Ke Fan, and Elena De Momi. Implementation and assessment of an augmented training curriculum for surgical robotics. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023 .
- [8] Francesca Fati, Marina Rosanu, Luigi De Vitis, and **Alberto Rota et al.** Deep learning for decision support in ovarian cancer treatment planning. *Nature Women's Health*, 2025 - Under Review .
- [9] Junling Fu*, **Alberto Rota***, Shufei Li, Jianzhuang Zhao, Qingsheng Liu, Elisa Iovene, Giancarlo Ferrigno, and Elena De Momi. Recent advancements in augmented reality for robotic applications: A survey. In *MDPI Actuators*, 2023 .

REFERRALS

I am grateful and proud to have worked alongside these people throughout my academic and professional career. Whether as advisors, mentors, or colleagues, they know my work closely.

- Advisor* **Prof. Elena De Momi** ✉ elena.demomi@polimi.it
PhD supervisor and head of NEARLab's Medical Robotics Section
- Mentor* **Mr. Stefano Pomati** ✉ stefano.pomati@gmail.com
PhD industry mentor and Director of Surgical Innovation at Asensus Surgical, now Karl Storz
- Host* **Prof. Benjamin Busam** ✉ b.busam@tum.de
Hosted my PhD research visit and heads of the Photogrammetry and Remote Sensing Lab at TUM

DISCLOSURES

- GDPR* I authorize the processing of personal data according to EU Regulation 679/2016 or according to the readers local regulations if not in the EU.
- Accessibility* I authorize the publication and the complete accessibility of this CV according to the italian D. Lgs n. 33 of March 14 2013.
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